「Def. A map $\mathbb{N}\times\mathbb{N}\to\mathbb{C} : (m,n)\mapsto a_{mn}$

is called a Double Sequence.

observe that: define a set $X \overset{\text{def}}{=} \{0\}\cup\{\frac{1}{n} \mid n\in\mathbb{N}\}$.
For a continuous map $f: X\to\mathbb{R}$, a sequence $a_n = f(\frac{1}{n})$ converges to $f(0)$. Conversely, for a convergent sequence $\{a_n\}$, the map f defined on X as

$$f(0) = \lim_{n\to\infty} a_n, \quad f(\tfrac{1}{n}) = a_n \quad (n\in\mathbb{N})$$

is continuous.

「Thm. Let $\{a_{m,n}\}$ be a double sequence. If $\{a_{mn}\}$ satisfies the followings:

1) For each $m\in\mathbb{N}$, $a_{mn}\to b_m$ as $n\to\infty$
2) For each $n\in\mathbb{N}$, $a_{mn}\to c_n$ as $m\to\infty$.
3) For any $\varepsilon>0$, there exists $N\in\mathbb{N}$ s.t. $m\geq N \Rightarrow \sup_{n\in\mathbb{N}} |a_{mn}-c_n|\leq\varepsilon$.

Then, $\lim_{m\to\infty} b_m = \lim_{m\to\infty}\lim_{n\to\infty} a_{mn} = \lim_{n\to\infty}\lim_{m\to\infty} a_{mn} = \lim_{n\to\infty} c_n$

Proof. For each $m\in\mathbb{N}$, define f_m on $\{\frac{1}{n}\mid n\in\mathbb{N}\}$ as $f_m(\frac{1}{n}) = a_{mn}$ $(n\in\mathbb{N})$
Then, we can observe that

1) $\Longleftrightarrow$ For each $m\in\mathbb{N}$, $\lim_{t\to 0} f_m(t) = b_m$
2) $\Longleftrightarrow$ f_m converges pointwise to f, where $f(\frac{1}{n}) = c_n$ $(n\in\mathbb{N})$
3) $\Longleftrightarrow$ f_m converges uniformly to f.

Therefore, $\lim_{m\to\infty} b_m = \lim_{m\to\infty}\lim_{t\to 0} f_m(t) = \lim_{t\to 0}\lim_{m\to\infty} f_m(t) = \lim_{t\to 0} f(t) = \lim_{n\to\infty} f(\frac{1}{n}) = \lim_{n\to\infty} c_n.$

Lemma. Let $f_n : X \to Y$ be continuous functions s.t $f_n \rightrightarrows f$.
If a sequence $\{x_n\}$ in X converges to $x \in X$, then $\{f_n(x_n)\}$ converges to $f(x)$.

Proof. Since $f_n \rightrightarrows f$ and f_n is continuous, f is continuous.
Given $\varepsilon > 0$, there exists $N_1 \in \mathbb{N}$ s.t $n \geq N_1 \Rightarrow \sup_X |f_n - f| < \frac{\varepsilon}{2}$.
Meanwhile, for these $n \geq N_1$, there is $N_2 \in \mathbb{N}$ s.t $m \geq N_2 \Rightarrow |f_n(x_m) - f_n(x)| < \frac{\varepsilon}{2}$.
Now, for $k \geq \max\{N_1, N_2\}$,

$$|f_k(x_k) - f(x)| \leq |f_k(x_k) - f_k(x)| + |f_k(x) - f(x)| < \varepsilon. \quad \blacksquare$$

Thm. Let $\{a_{mn}\}$ be a double sequence s.t

1) For each $m = 1, 2, \cdots$, $\sum_{n=1}^{\infty} |a_{mn}| = b_m$

2). $\sum_{m=1}^{\infty} b_m$ converges.

Then,

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{mn} = \lim_{N\to\infty} \sum_{m,n=1}^{N} a_{mn}. \quad \cdots (*)$$

Proof. Let $X = \{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{N}\}$, and for each $m \in \mathbb{N}$, define

$$f_m : X \to \mathbb{R} ; x \mapsto \begin{cases} \sum_{j=1}^{\infty} a_{mj} & \text{if } x = 0 \\ \sum_{j=1}^{n} a_{mj} & \text{if } x = \frac{1}{n},\ n \in \mathbb{N}. \end{cases} \quad \left(\text{That is, } f_m\left(\tfrac{1}{n}\right) = \sum_{j=1}^{n} a_{mj}\right)$$

Weierstrass M-test

and let $g_n = \sum_{m=1}^{n} f_m$. Since $|f_m| \leq \sum_{j=1}^{\infty} |a_{mj}| = b_m$ and $\sum b_m$ converges, g_n converges uniformly.
And, each f_m is continuous, thus $g_n = \sum_{m=1}^{n} f_m$ is continuous, thus $g = \lim_{n\to\infty} g_n$ is continuous, therefore

$$\underbrace{\lim_{n\to\infty} g\left(\tfrac{1}{n}\right) = g(0)}_{g\text{-conti.}} \underbrace{= \lim_{n\to\infty} g_n\left(\tfrac{1}{n}\right)}_{\text{Lemma.}}$$

. Moreover,

$$\lim_{n\to\infty} g\left(\tfrac{1}{n}\right) = \lim_{n\to\infty} \sum_{m=1}^{\infty} f_m\left(\tfrac{1}{n}\right) = \lim_{n\to\infty} \sum_{m=1}^{\infty} \sum_{j=1}^{n} a_{mj} = \lim_{n\to\infty} \sum_{j=1}^{n} \sum_{m=1}^{\infty} a_{mj} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{mn}$$

$$g(0) = \sum_{m=1}^{\infty} f_m(0) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn}$$

In finite sum, convergents preserved limit.

$$\lim_{N\to\infty} g_N\left(\tfrac{1}{N}\right) = \lim_{N\to\infty} \sum_{m=1}^{N} f_m\left(\tfrac{1}{N}\right) = \lim_{N\to\infty} \sum_{m=1}^{N} \sum_{j=1}^{N} a_{mj} = \lim_{N\to\infty} \sum_{m,n=1}^{N} a_{mn}.$$

$\blacksquare$

Note: If $(*)$ holds then we say the sum of $\{a_{mn}\}$ can be exchanged.

「Thm. If $\{a_{mn}\}$ is positive, then sum of $\{a_{mn}\}$ can be exchanged.

Proof. For any $M, N \in \mathbb{N}$ s.t. $M \le N$, since $a_{mn} \ge 0$,

$$\sum_{m=1}^{M}\sum_{n=1}^{N} a_{mn} \le \sum_{m,n=1}^{N} a_{mn}$$

Taking limit to the both side, then

$$\sum_{m=1}^{M}\sum_{n=1}^{\infty} a_{mn} = \lim_{N\to\infty}\sum_{m=1}^{M}\sum_{n=1}^{N} a_{mn} \le \lim_{N\to\infty}\sum_{m,n=1}^{N} a_{mn}$$

Since M was arbitrary,

$$\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} a_{mn} \le \sum_{m,n=1}^{\infty} a_{mn}.$$

And, the converse inequality is given by: for any $N \in \mathbb{N}$,

$$\sum_{m,n=1}^{N} a_{mn} \le \sum_{n=1}^{N}\sum_{m=1}^{\infty} a_{mn}.$$

「Corollary. If $a_k \ge 0$ for all $k \in \mathbb{Z}$, then

$$\lim_{n\to\infty}\sum_{k=-n}^{n} a_k = \lim_{n\to\infty}\lim_{m\to\infty}\sum_{k=-m}^{n} a_k = \lim_{m\to\infty}\lim_{n\to\infty}\sum_{k=-m}^{n} a_k.$$